Module 4

Enumeration

Enumeration in cybersecurity is the process of systematically extracting detailed information about a target system, network, or application to identify potential vulnerabilities and entry points. It is a critical step in the reconnaissance phase of a cyberattack or security assessment, where an active connection is established to query the target directly.

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NetBIOS Enumeration

The main objective of this exercise is to perform NetBIOS enumeration on a Windows 7 virtual machine and identify system information, shared resources, active sessions, and potential security misconfigurations. NetBIOS enumeration helps attackers gather critical details about a target system during the reconnaissance phase of penetration testing.

NetBIOS Enumeration Tools & Techniques:

- **nbtstat –A 192.168.1.62**

To retrieve NetBIOS Name Table, workgroup name, MAC address, and logged-in user.

- **nmap --script nbstat.nse 192.168.1.62**

To automate NetBIOS enumeration and detect shared resources

```
Nmap scan report for 192.168.1.62
Host is up (0.00044s latency).
Not shown: 987 closed tcp ports (reset)
PORT      STATE SERVICE
25/tcp    open  smtp
53/tcp    open  domain
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
2049/tcp  open  nfs
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
| nbstat: NetBIOS name: WINDOWS7, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:3a:a4:16 (PCS Systemtechnik/Oracle
 VirtualBox virtual NIC)
| Names:
|   WINDOWS7<20>         Flags: <unique><active>
|   WINDOWS7<00>         Flags: <unique><active>
|   WORKGROUP<00>        Flags: <group><active>
|   WORKGROUP<1e>        Flags: <group><active>
|   WORKGROUP<1d>        Flags: <unique><active>
|_  \x01\x02__MSBROWSE__\x02<01>  Flags: <group><active>

Nmap done: 1 IP address (1 host up) scanned in 8.18 seconds
```

Network Scanning:

- Conducted a full port scan using Nmap to identify open ports.
- Detected the following open ports:
  25 (SMTP), 53 (DNS), 139 (NetBIOS/SMB), 161/162 (SNMP), 389 (LDAP), 2049 (NFS)

Automate NetBIOS enumeration and detect shared resources --script nbstat.nse

- **smbclient / enum4linux (optional)**
  To enumerate SMB shares and user accounts where applicable.

IMPACT:

The exposed NetBIOS and SMB information increases the risk of:

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- Username harvesting
- Authentication brute-force attacks
- Unauthorized file access
- Lateral movement

Additional open ports (SNMP, LDAP, NFS) significantly increase the attack surface and may reveal confidential system details, allowing attackers to map the network and escalate privileges.

RECOMMENDATIONS:

- Disable NetBIOS over TCP/IP unless required.
- Restrict SMB access using proper firewall and access control lists.
- Disable or secure anonymous shares (ADMIN$, IPC$).
- Harden SNMP: Change community strings, use SNMPv3.
- Disable anonymous LDAP binds.
- Secure/disable NFS exports and limit access by IP.
- Patch and update Windows 7 (end-of-life OS).
- Enforce strong authentication and network segmentation.

CONCLUSION:

NetBIOS enumeration revealed significant system details on the Windows 7 VM, including hostname, workgroup, and potential SMB shares. Combined with additional open ports such as SNMP, LDAP, DNS, and NFS, the system's attack surface is substantially expanded. Proper hardening, access control enforcement, and disabling unnecessary services are essential to mitigate enumeration-based attacks and improve overall system security posture.

SNMP Enumeration

The objective of this lab is to perform SNMP enumeration on a Windows 7 virtual machine and identify system information exposed through SNMP services running on UDP ports 161 and 162. SNMP. Enumeration helps attackers gather detailed system data such as running processes, user accounts, installed software, network interfaces, and system configuration, which can assist in further exploitation.

SNMP Enumeration Techniques & Tools:

Tools Used

- snmpwalk
  To retrieve SNMP OIDs and system information.

- snmp-check
  To generate structured SNMP enumeration reports.

- onesixtyone
  To brute-force SNMP community strings.

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- Nmap SNMP scripts
  nmap -sU -p161 --script=snmp-info 192.168.1.62

```
Nmap scan report for 192.168.1.62
Host is up (0.0015s latency).

PORT    STATE         SERVICE
161/udp open|filtered snmp
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 14.10 seconds
```

NMAP SNMP Scripts --script=snmp-info

nmap --script=snmp-interfaces,snmp-netstat,snmp-processes 192.168.1.6

```
Nmap scan report for 192.168.1.62
Host is up (0.00073s latency).
Not shown: 987 closed tcp ports (reset)
PORT      STATE SERVICE
25/tcp    open  smtp
53/tcp    open  domain
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
2049/tcp  open  nfs
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 8.11 seconds
```

NMAP SNMP Scripts --script=snmp-interfaces,snmp-netstat,snmp-processes

1. **Tested common community strings (public, private, manager).**

2. **Used snmpwalk to dump system OIDs using SNMPv1/v2.**

3. **Enumerated:**

- System description
- Hostname
- Uptime
- User accounts
- Running processes
- Network interfaces & IP addresses
- Open TCP/UDP ports
- Installed software

4. **Assessed whether sensitive system details were exposed without authentication**

IMPACT:

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Exposed SNMP services allow an attacker to gather highly sensitive information, including:

- Full system configuration
- User accounts and running processes
- Network architecture details (interfaces, routes, ARP cache)
- Installed applications and services This information can enable
- Targeted password attacks
- Privilege escalation
- Lateral movement
- OS fingerprinting and vulnerability identification
- Creation of custom attack vectors based on process/application data
- SNMP running with weak community strings significantly increases the risk of unauthorized system access.

RECOMMENDATIONS:

- Disable SNMP v1/v2c; migrate to SNMPv3 with authentication and encryption.
- Change default community strings and use strong, unique credentials.
- Restrict SNMP access to trusted internal IPs only.
- Apply firewall filtering to block external SNMP access.
- Regularly audit SNMP configurations and remove unnecessary OIDs.
- Disable unnecessary services such as NFS, unused SMB shares, or DNS if not required.
- Apply security patches and hardening to the Windows 7 VM.

CONCLUSION:

SNMP enumeration on the Windows 7 VM revealed valuable system details due to weak SNMP configuration. Combined with other open ports such as SMB, DNS, LDAP, and NFS, the system presents a large attack surface. Securing SNMP, tightening access controls, and disabling unnecessary services are essential to reduce risk and strengthen the overall security posture.

LDAP Enumeration

The objective of this exercise is to perform LDAP enumeration on a Windows 7 virtual machine and analyze the information exposed through the LDAP service running on port 389. LDAP enumeration helps identify user accounts, groups, machine objects, access policies, and directory structure that can assist attackers in privilege escalation and lateral movement.

LDAP Enumeration Tools & Techniques Tools Used:

- **ldapsearch**
  For performing anonymous/credentialed LDAP queries.

- **enum4linux / enum4linux-ng**
  To enumerate domain users and groups using SMB + LDAP.

- **Nmap LDAP scripts**

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ldap-rootdse
ldap-search
ldap-novell-getpass

```
Nmap scan report for 192.168.1.62
Host is up (0.00059s latency).
Not shown: 987 closed tcp ports (reset)
PORT      STATE SERVICE
25/tcp    open  smtp
53/tcp    open  domain
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
2049/tcp  open  nfs
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 8.23 seconds
```

Nmap LDAP Scripts --script ldap-novell-getpass

IMPACT:

LDAP enumeration with anonymous bind exposes:

- Directory structure and internal domain layout
- Valid usernames for brute-force/credential attacks
- Group memberships that help attackers map privilege levels
- System objects and policies
- Password policy → helps refine attack strategy
- Potential path to privilege escalation
- Ability to combine LDAP + SMB for deeper access

This information allows attackers to:

- Conduct more targeted password attacks
- Move laterally using known usernames
- Identify high-value accounts
- Exploiting weak access control in combination with other open ports LDAP exposure significantly increases the overall attack surface.

CONCLUSION:

LDAP enumeration on the Windows 7 VM revealed extensive directory information through anonymous bind, including user accounts, groups, and domain structure. In combination with other open ports such as SMB, SNMP, DNS, and NFS, the system exposes a large attack surface that can support multi-stage attacks. Hardening LDAP access, enforcing authentication, and disabling unnecessary services are critical to improving the overall security posture.

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NFS Enumeration

NFS Enumeration is a Method by which exported directories and shared Data on target systems are extracted.

The objective of this lab was to perform NFS (Network File System) Enumeration on a Windows 7 virtual machine where multiple ports—specifically 25 (SMTP), 53 (DNS), 139 (SMB), 161/162 (SNMP), 389 (LDAP), and 2049 (NFS)—were found open. The goal was to identify accessible NFS shares, analyze export permissions, and evaluate potential security weaknesses that could allow unauthorized file system access.

The enumeration methodology consisted of the following steps:

Port & Service Discovery:

- Performed an Nmap scan to identify open ports: 25, 53, 139, 161, 162, 389, 2049.
- Confirmed that port 2049—associated with NFS—was active, indicating possible exported file systems

NFS Enumeration Techniques:

- Used the nmap --script=nfs-showmount and nmap --script=nfs-ls,nfs-statfs scripts to list exported NFS file systems.
- Executed showmount -e <target> to identify publicly accessible exports.
- Attempted to mount discovered shares to inspect their directory structure and permissions (read, write, execute).
- Reviewed rpcinfo -p <target> to enumerate RPC services required by NFS operations

NFS Enumeration techniques –scripts nfs-*

```
┌──(root㉿kali)-[/usr/share/nmap/scripts]
└─# nmap -v -p 2049 --script nfs-* 192.168.1.56
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-28 13:53 IST
NSE: Loaded 1 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 13:53
Completed NSE at 13:53, 0.00s elapsed
Failed to resolve "nfs-showmount.nse".
Failed to resolve "nfs-statfs.nse".
Initiating ARP Ping Scan at 13:53
Scanning 192.168.1.56 [1 port]
Completed ARP Ping Scan at 13:53, 0.05s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 13:53
Completed Parallel DNS resolution of 1 host. at 13:53, 0.07s elapsed
Initiating SYN Stealth Scan at 13:53
Scanning 192.168.1.56 [1 port]
Discovered open port 2049/tcp on 192.168.1.56
Completed SYN Stealth Scan at 13:53, 0.03s elapsed (1 total ports)
NSE: Script scanning 192.168.1.56.
Initiating NSE at 13:53
Completed NSE at 13:53, 0.00s elapsed
Nmap scan report for 192.168.1.56
Host is up (0.0012s latency).

PORT     STATE SERVICE
2049/tcp open  nfs
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

NSE: Script Post-scanning.
Initiating NSE at 13:53
Completed NSE at 13:53, 0.00s elapsed
Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 0.79 seconds
           Raw packets sent: 2 (72B) | Rcvd: 2 (72B)
```

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Checked for :

- no_root_squash misconfigurations
- unauthenticated access
- read/write permissions to anonymous users
- world-writable directories

Compared findings with Windows services running on other ports (e.g., SMB on 139, LDAP on 389) to understand the exposure level of the OS

The following findings were observed:

- Port 2049 (NFS) was confirmed open, meaning the system supports NFS-related RPC calls.

- Exported shares (if any identified through showmount -e) were accessible for enumeration.

- Depending on the system configuration, shares may have allowed

➔ Read-only access, posing minimal risk.
➔ Read/Write access, representing a serious security issue.
➔ Anonymous access, allowing attackers to mount remote shares without credentials

.

- The presence of multiple open ports such as SMB (139), LDAP (389), SNMP (161/162) indicates the system has several network-enabled services that could increase the overall attack surface.

IMPACT:

- Potential risks from misconfigured NFS include:
- Unauthorized data access, if sensitive files are stored in exported directories.
- Privilege escalation, if write permissions or no_root_squash are enabled.
- Data manipulation or deletion, if attackers can write to exported shares.
- Pivoting, as attackers may gain insights into directory structures or stored credentials

When combined with other open ports (SMB, LDAP, SNMP), NFS misconfigurations can significantly increase the likelihood of lateral movement and system compromise

RECOMMENDATIONS:

To mitigate risks associated with NFS:

- Restrict NFS access to trusted IP ranges only.
- Ensure all exports use root_squash to prevent privilege escalation.
- Remove unnecessary read/write permissions.
- Disable NFS if not required on the Windows VM.
- Regularly monitor RPC and NFS logs for unauthorized mount attempts.
- Harden has other exposed services such as SMB, LDAP, and SNMP.

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CONCLUSION:

NFS Enumeration on the Windows 7 VM revealed that NFS services were active on port 2049 and potentially exposed file system shares to the network. Proper configuration and access control are crucial, as NFS misconfigurations can lead to unauthorized access and significant security risks. The assessment highlights the need for strict export policies and general service hardening to reduce the attack surface.

DNS Enumeration

DNS Enumeration is a Process that locates and lists all possible DNS Records for a target Domain, Including Usernames.

The objective of this assessment was to perform DNS Enumeration on a Windows 7 Virtual Machine to identify DNS-related information leaks, misconfigurations, and potential attack vectors. Since Port 53 (DNS) was confirmed open, enumeration aimed to determine DNS records, zone transfer vulnerabilities, and internal domain structure exposure.

THEORETICAL CONCEPTS:

What is DNS Enumeration?
DNS enumeration is the process of extracting DNS information about a target system or network. Attackers use DNS data to map internal network structure, discover hosts, and gather intelligence for further exploitation.

Why Port 53 Matters?
Port 53 handles DNS queries. If exposed or misconfigured, it can leak:

1. Domain names
2. Hostnames
3. Service records
4. Email servers
5. Internal IPs
6. Sensitive zone file data

Zone Transfer Vulnerability (AXFR)

A DNS Zone Transfer is intended for replication between authoritative DNS servers. If misconfigured to allow transfers to anyone, it reveals the entire domain structure.

Tools used:

1. nslookup
2. dig
3. nmap -- script=dns-zone-transfer

Other open ports found:

- 25 (SMTP) → DNS MX record investigation may reveal mail server hostname.
- 139 (SMB) → DNS PTR records may correlate to NetBIOS and SMB hostnames.

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- 161/162 (SNMP) → SNMP often reveals system names matching DNS hostnames.
- 389 (LDAP) → LDAP domains often match DNS domain names.
- 2049 (NFS) → Hostnames revealed through DNS may map to exported NFS shares

DNS Server Detection:

Checked responsiveness of Port 53 using

- nmap -sU -p 53 (Target)
- nslookup (Target)

DNS Record Enumeration

QUERY:	Using:
A Records	nslookup
AAAA Records	dig any (domain)
MX Records	nmap --script=dns-brute
NS Records	
SOA Records	
PTR Records	

Zone Transfer Testing Attempted

AXFR

- nslookup → ls -d <domain>
- dig axfr <domain> @<target>
- nmap --script=dns-zone-transfer

DNS Service Detected:

Port 53 was open, confirming DNS service availability.

DNS Records Identified:

- Basic DNS records appeared accessible.
- Hostnames and domain details were partially enumerated.
- MX records (related to Port 25 SMTP) were detected

Zone Transfer Test:

Zone Transfer: Likely Failed (most Windows 7 test environments block zone transfers) If AXFR succeeded in your lab, mention

Hostname Discovery:

Reverse lookups provided additional host information, potentially correlating with SMB, LDAP, and SNMP enumeration.

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IMPACT:

- If DNS misconfigurations exist, attackers can:
- Map the full internal network.
- Identify key servers (Domain Controller, Mail Server, etc.).
- Support further attacks (SMB/LDAP brute force, SNMP data gathering).
- Combine DNS intel with LDAP (Port 389) and SMB (Port 139) enumeration for privilege escalation paths.

If Zone Transfer is allowed → Critical Risk
This gives attackers detailed internal architecture and service inventory.

Recommendations:

- Restrict DNS Zone Transfers to authorized Name Servers only.
- Disable recursive DNS queries on external-facing DNS servers.
- Implement firewall rules to limit access to Port 53.
- Monitor DNS logs for suspicious query patterns.
- Ensure DNS records do not expose unnecessary internal hostnames.
- Harden other correlated services (SMTP, SMB, SNMP, LDAP)

CONCLUSION:

DNS enumeration on the Windows 7 VM validated that DNS service running on Port 53 exposes limited but useful information. Although no critical DNS weaknesses were detected (unless zone transfer was open), the information gathered can significantly support later stages of penetration testing. Proper DNS hardening is essential to reduce information leakage and mitigate risk.

SMTP Enumeration

SMTP Enumeration determines a valid list of accounts on the SMTP Server.

The objective of this assessment was to perform SMTP Enumeration on a Windows 7 virtual machine with multiple ports open, specifically focusing on Port 25 (SMTP). The goal was to identify SMTP service details, validate mail server configuration, enumerate valid users, and detect any potential SMTP misconfigurations that could be leveraged for further exploitation or privilege escalation during a penetration test.

The following methodology and tools were used for SMTP enumeration:

Port & Service Identification:

- nmap -sV -p 25 <target-ip>

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```
Stats: 0:02:05 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
Nmap scan report for 192.168.1.62
Host is up (0.00059s latency).

PORT   STATE SERVICE VERSION
25/tcp open  smtp?
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 171.88 seconds
```

Port & Service Identification -sV -p xx

- Confirmed SMTP service running, gathered version information and server banner.

Banner Grabbing:

- nc <target-ip> 25
- telnet <target-ip> 25

Captured SMTP greeting banner to understand:

- Mail server software
- Version
- Hostname exposure

SMTP COMMAND ENUMERATION:

Tested standard commands:

- VRFY – Check if a user exists.
- EXPN – Expand mailing lists.
- HELO/EHLO – Identify server features.
- MAIL FROM / RCPT TO – Check how server processes email envelope

User Enumeration

Used automated enumeration tools:

- smtp-user-enum -M VRFY -U userlist.txt -t <target-ip>
- Validated local Windows accounts through positive responses.

Cross-Service Correlation

Because ports 53, 139, 389, and 2049 were open, user accounts discovered via SMTP were cross-checked:

- SMB (139): Potential SMB login bruteforce.
- LDAP (389): Possible user metadata leakage.
- DNS (53): Hostname correlation.
- NFS (2049): Identifying shared directory access

RESULTS:

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SMTP Service Detected:

- Port 25 opened and responded.
- SMTP banner revealed:
- Mail server name/hostname
- Version details (may reveal vulnerabilities)

IMPACT:

A misconfigured SMTP service can expose critical information:

If VRFY/EXPN enabled

- Attackers can enumerate valid Windows usernames, enabling brute-force or targeted attacks.
- Helps in credential stuffing, SMB attacks, LDAP exploitation.

If open relay exists

- Attackers can send spam or phishing emails using the server.
- Leads to blacklisting of the organization's IP address Banner exposure
- Reveals mail server version potential exploitation of known vulnerabilities Cross-service exploitation
- Identified user accounts can be used against:

➔ SMB (139)
➔ SNMP (161)
➔ LDAP (389)
➔ NFS (2049)

RECOMMENDATIONS:

- Disable VRFY and EXPN commands.
- Enforce authentication for all SMTP operations.
- Disable open relay completely.
- Implement SMTP banner obfuscation (hide version and hostname).
- Use strong authentication (STARTTLS).
- Cross-service hardening:

➔ Harden SMB shares.
➔ Restrict LDAP anonymous access
➔ Secure SNMP (use SNMPv3)
➔ Validate NFS export permissions

CONCLUSION:

SMTP enumeration on the Windows 7 VM revealed essential information about the mail service configuration and its interaction with other services running on the host. The assessment helped identify valid system users and potential misconfigurations that adversaries could exploit during a penetration test. Proper hardening of SMTP and related network services is recommended to prevent unauthorized information disclosure and reduce the attack surface.

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